SOFTWARE TEST LOG AND VERIFICATION RECORD

Project: Musanze Harvest Intelligence Service

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Test environment: Windows PowerShell; Python virtual environment; pytest; FastAPI

1. Test scope

The verification covers reproducible dataset generation, input validation, trained-model inference, API behavior and automated software checks. The entries below reflect the command outputs and final implementation results supplied for this report.

2. Command-line and API test results

ID	Test case and observed result	Result
CLI-01	Valid farm JSON returned a harvest estimate, risk class, probability, cluster and verification code.	PASS
CLI-02	Invalid record reported missing required fields; process exit code was 1.	PASS
API-01	GET /health returned status: healthy.	PASS
API-02	GET /model-info returned AI-A2-7341 and three model categories.	PASS
API-03	POST /predict accepted valid JSON and returned prediction values.	PASS
API-04	POST /predict rejected rainfall_mm=2000 with HTTP 422.	PASS
RUN-01	Full pipeline completed with institutional username and seed 7341.	PASS
RUN-02	Dataset SHA-256 fingerprint remained consistent across repeated generation.	PASS

3. Automated testing

The initial command-line pytest execution returned 6 passed, 1 warning in 5.22 seconds. The updated implementation test summary reports 8 passed, 1 warning. The warning concerns a Starlette/httpx deprecation and was not reported as a test failure.

The final test suite covered identity and seed validation, reproducibility, shared test-set consistency, confusion matrix structure, saved-model inference and API request validation.

4. Data and model verification

Check	Recorded outcome
Data generation	242 base rows; 3 added duplicate records; 245 raw rows; 242 after deduplication.
Missing values	12 missing soil moisture cells and 13 missing fertilizer cells in raw CSV.
SHA-256	408c124768bd6ff6bb83987a961dff1077a34ee08873b70363626f95b57e5fa5

Regression	LinearRegression; final test MAE 0.7048; RMSE 0.8876.
Classification	RandomForestClassifier; precision 0.9259; recall 1.0000; confusion matrix [[22, 2], [0, 25]].
Clustering	K-Means selected K=2; silhouette score 0.1576, indicating weak cluster separation.
Model files	Regression, classification and clustering saved pipelines were reported to load successfully.

5. Interpretation and conclusion

The recorded outcomes show successful validation of tested commands and requests, including appropriate rejection of invalid inputs. The results demonstrate operation on a synthetic farm dataset; they do not establish accuracy on real farms. The final test count is taken from the updated test execution summary and should be matched to the local pytest output during submission review.